ENGLISH PROJECT REPORT VERSION

Drone Project - Second Progress Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Progress report
Primary Role	Team lead and integration planner
Original Source	drone-second-progress-report.pdf (24 pages)

Executive Summary

The second progress stage moved the project from concept to early flight and ball-handling tests. The team began seeing how design assumptions changed once hardware was assembled.

The report documents revision of the ball-handling concept and the shift from ideal design to integration reality.

Applicant Contribution Signals

- Responded to early flight-test evidence rather than defending the first concept.
- Adjusted mechanical direction based on stability, payload behavior, and fabrication limits.
- Kept the team aligned as design choices became more constrained.

Build-Test Revision

Early tests exposed the interaction between payload design and drone stability.

The team revised the ball-handling approach to reduce integration risk and improve final feasibility.

Engineering Learning

Physical testing made clear that lightweight design, control margin, and assembly quality had to be developed together.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Early test mismatch	Revised ball-handling approach	Adapted to real build behavior	Iterative design

Constraint	Decision	Why It Mattered	Skill Demonstrated
Payload stability	Changed integration assumptions	Reduced flight risk	System debugging
Schedule pressure	Focused on feasible next prototypes	Protected final progress	Leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.